

Jiaheng Li

Full Stack Software Engineer

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[Portfolio](#) · [GitHub](#) · Auckland, NZ

Open Work Visa to 25 Feb 2029 · No sponsorship required

Available immediately · Open to relocate

At a Glance

Experience: 3 years commercial development

Data: SQL · PostgreSQL · Oracle

Systems: Security · Distributed computing · FFmpeg

Languages: Go · Python · Java · C# · C/C++

Backend: Go/Python Web APIs · REST · .NET

Frontend: React · TypeScript · HTML/CSS/JS

Professional Experience

Skyline Consulting Engineers Limited

Oct 2024 - Dec 2025

Part-time Full-stack Developer · Auckland

Turned structural engineers' calculation workflows into C#/.NET, React/TypeScript and PostgreSQL software. Clarified requirements with engineers and checked calculation results against analytical benchmarks.

Yonyou Network Technology Co., Ltd.

Mar 2022 - Dec 2023

Java Backend Developer · Zhengzhou, China

Developed Java/Spring REST APIs and accounting forms for a multi-tenant financial SaaS ERP. Worked with Oracle, PostgreSQL and MyBatis, and built cross-module login and Kafka message processing for workloads above 100,000 records.

Projects

[Go Kademia DHT](#)

Go · DHT · P2P
Concurrent networking

Implemented four Kademia RPCs in Go, with XOR-distance routing and concurrent queries. Integrated the DHT with a torrent-download session; documented peer discovery, packet captures and a successful torrent download in my undergraduate thesis.

[Blockchain PKI](#)

Blockchain · PKI
Network security

Explored lightweight blockchain-based certificate issuance, validation and revocation, including smart-contract concepts; produced a COMPSCI 726 research report.

[P2P Lockstep Kit](#)

P2P protocol · WebRTC
WebSocket · State sync
TypeScript

Built a browser-game toolkit with separate network, session and UI packages. WebSocket signalling connects peers over WebRTC; sessions handle ordered moves, Ready/Start, undo/restart and reconnect synchronisation. Web Components and a game-plugin interface separate rules from networking. Powers four live games, including multiplayer Mahjong, with an AI Skill for building new games.

Live demos: [Mahjong](#) · [Gomoku](#) · [Chess](#) · [Xiangqi](#)
npm packages: [Network](#) · [Session](#) · [UI](#) · [AI build Skill](#)

[NX-Cast](#)

C/C++ · TCP/IP · DLNA
Embedded networking

Turned a Nintendo Switch into a media receiver supporting DLNA, IPTV and AirPlay 2. Cross-compiled and integrated FFmpeg/libmpv, using event loops, multithreading and state machines to manage playback within embedded memory and CPU constraints.

[Total-Order Multicast](#)

C# · Distributed protocols
TCP · Logical clocks

Implemented and tested consistent message-delivery ordering across five processes under simulated network delays for a COMPSCI 711 assignment.

Education

Master of Information Technology · University of Auckland · Nov 2025

Completed COMPSCI 726: Network Defence and Countermeasures, and COMPSCI 711: Parallel and Distributed Computing.

Bachelor of Network Engineering · Henan University · Jun 2022

Former employers are willing to provide references.